PhD Dissertation Proposal Defense: Written Report

From Prompt Wrangling to Retrieval vs. Abstraction:
A Two-Study Investigation of LLM Generalization
in Physics-Driven Procedural Content Generation

Arash Moradi Karkaj
PhD Candidate
New Jersey Institute of Technology
am3493@njit.edu

Ioannis Koutis
Advisor
New Jersey Institute of Technology
ikoutis@njit.edu

Abstract

This report synthesizes two related studies that use the ChatGPT4PCG competition as a controlled laboratory for probing generalization in large language models (LLMs). The first study (*Prompt Wrangling*, FDG 2024) establishes replication baselines and identifies a fundamental tension between competitive performance and genuine generalization when a single model, GPT-3.5, is evaluated across prompt engineering strategies. This work was presented as the Qualification Exam for the PhD Candidate.

The second expanded study, pondering on the feedback received after successful QE, extends the benchmark to 25 LLMs spanning seven developer families and introduces a contiguous-coverage metric based on longest-common-substring analysis to distinguish verbatim retrieval from abstractive transfer. Together, the two studies argue that high task scores on this benchmark are predominantly driven by memorization of in-context examples rather than learned generalization, a finding that holds across model families, scales, and post-training regimes.

1 Background and Motivation

The ChatGPT4PCG competition [1] tasks participants with crafting a single prompt template of at most 900 words that instructs an LLM to generate Angry Birds game levels shaped like each of the 26 uppercase English letters. Levels are assembled by issuing a series of `ab_drop(block_type, x_position)` commands, which are then converted to XML and simulated in the Unity-based *Science Birds* physics engine. Each generated level is scored on two independent axes: **stability** (fraction of blocks that do not move during a 10-second simulation) and **similarity** (softmax probability assigned by a fine-tuned Vision Transformer (ViT) to the target letter). A per-prompt score is the unweighted mean of stability$\times$similarity products across all 260 generated levels (10 per letter, 26 letters). You can see an overview of the pipeline in Figure 1.

The competition received 18 submissions, of which 15 qualified after checking for length ($\leq$900 words), permitted character set, and the presence of the `<OBJECT>` placeholder token. The original competition used GPT-3.5 as the sole generation model. This controlled setting makes the benchmark uniquely suited to studying generalization: the task lies outside standard NLP evaluation regimes, the output is procedural rather than natural-language, correctness is determined only after physics simulation and visual classification, and the submitted prompts vary systematically in the number of explicit letter solutions they embed (from 0 to 26).

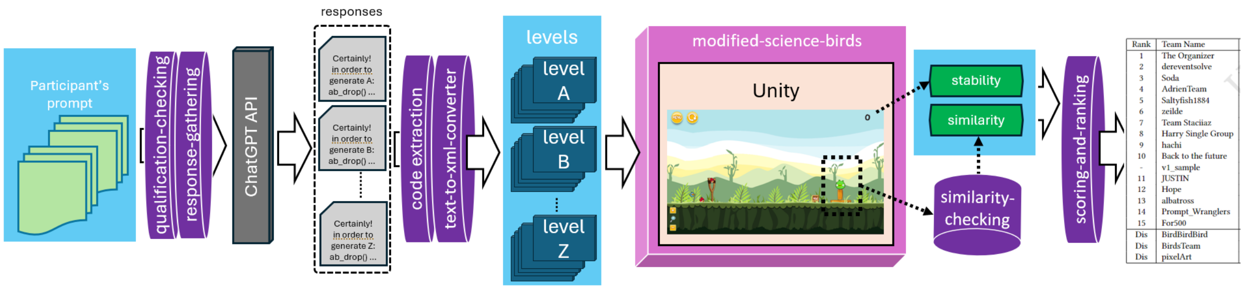
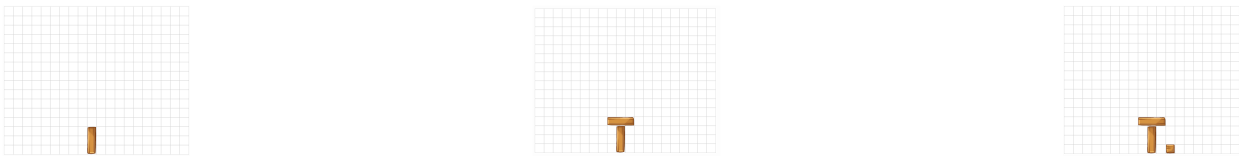


Figure 1: The simplified original ChatGPT4PCG competition pipeline.



(a) Step 1: One-by-Three ab_drop('13',9)

(b) Step 2: Three-by-One ab_drop('31',9)

(c) Step 3: One-by-One ab_drop('11',11)

Figure 2: Three block drops.

An example of building the letter 'T' is shown in Figure 2. The generator (a) first drops a one-by-three brick at $x = 9$, then (b) drops a three-by-one brick at $x = 9$, which stacks on the previous brick. Generators drop these blocks through the drop function *ab_drop(block_type, position)*, where the first argument is one of the three available block shapes, and the second argument is the x-position of the drop. Besides the two bricks used to construct this 'T' shape, the third brick available is a one-by-one square, shown dropped next to the 'T' in (c).

2 Study I – Prompt Wrangling (FDG 2024 [2])

2.1 Replication Experiment

The first contribution of Study I is a replication of the original competition results, run approximately six months after the official evaluation and on independent hardware. Each of the 15 qualified prompts was re-queried against GPT-3.5 with both n=10 and n=100 responses per letter, yielding 260 and 2,600 generated levels per prompt respectively.

Sample-size adequacy. Rankings obtained at n=10 and n=100 are identical, confirming that 10 responses per letter is statistically sufficient to rank prompts reliably given the stochasticity of the domain.

Temporal drift of the model. Despite using the same scripts and prompts, two submissions deviated substantially from their original standings. *Prompt_Wranglers* rose from rank 14 (score 0.00) to rank 3 (score 14.34), and *hachi* rose from rank 9 (score 1.57) to rank 4 (score 13.11). For *Prompt_Wranglers*, the improvement was traced to a change in GPT-3.5's formatting behavior: at the time of the original competition, the model omitted the triple-backtick code delimiters required by the extraction script, producing invalid output; by replication time, the model had begun including them consistently. The cause of *hachi*'s improvement could not be fully determined, underscoring a

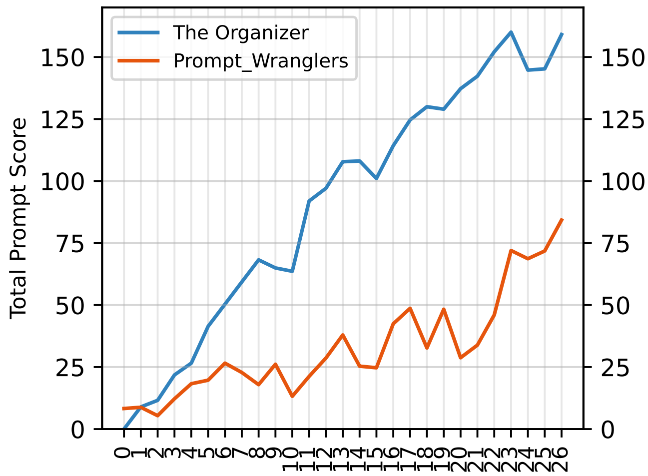


Figure 3: N-shot prompting experiments, where N ranges from 0 at the left to 26 at the right. These experiments use both 'Prompt_Wranglers' and 'The Organizers' prompt as starting points. Scores are raw total scores.

broader limitation: experiments conducted against closed-source, continuously updated APIs are inherently difficult to reproduce without archiving raw model outputs.

2.2 Generalization Experiment

The second contribution is a systematic generalization study. The key observation motivating it is that the top-performing prompt in the original competition (*The Organizer*) embeds explicit block-drop sequences for all 26 letters, effectively providing the model with a lookup table rather than requiring it to generalize structural knowledge. This is at odds with the PCGML goal of generating novel, previously unseen content from learned representations [3].

Three experimental regimes were evaluated:

- **Zero-shot**: All explicit letter solutions removed from each prompt.
- **One-shot**: One example letter retained; 26 variants tested per prompt.
- **N-shot**: Number of embedded solutions varied from 0 to 26 for two prompts (*The Organizer* and *Prompt_Wranglers*).

Results. Performance degrades sharply under all restricted settings. The best zero-shot prompt achieves a raw score of only 0.074, roughly 12% of the top replicated score (0.605 for *The Organizer* at full shots). The N-shot curve rises approximately linearly with the number of provided examples for both tested prompts, implying that performance tracks example count rather than genuine transfer—providing twice as many explicit solutions approximately doubles the score. Rankings in the zero-shot regime differ substantially from competition rankings, suggesting that prompt quality as measured by the competition metric conflates lookup-table fidelity with generalization capacity.

Conclusion of Study I. GPT-3.5 is memorizing from in-context examples rather than abstracting structural rules for letter construction. The competition metric rewards completeness of the lookup table, not generalization. This finding motivates extending the benchmark to a broader model landscape with explicit memorization-detection tooling.

3 Study II – Retrieval vs. Abstraction Across 25 LLMs

3.1 Extended Model Landscape

Study II retains the 15 original human-authored prompts (plus one machine-authored prompt generated by Claude Sonnet 4.5 for itself) and evaluates all of them across 25 LLMs from seven developer families: Google (Gemma 3), Meta (Llama 3/3.1/3.2), Qwen (Qwen 2.5, Qwen 3), Mistral AI (Mistral-7B v0.1–v0.3), Microsoft (Phi-4 variants), DeepSeek AI (R1 distilled variants), and Anthropic (Claude Sonnet 4.5). The roster includes base, instruction-tuned, reasoning-tuned, and distilled variants, yielding a total of approximately 89,700 raw responses. All models were loaded locally via the HuggingFace `transformers` library and run at developer-default temperatures. The full model collection is listed in Table 1.

Table 1: Model collection specifications, categorised by developer. *Category* abbreviations: **B** = base, **I** = instruct, **R** = reasoning, **D** = distilled. *Updated* refers to the date the authors loaded each model for experiments.

Developer	Model	Params	Cat.	Context	Temp.	Updated
Google	gemma-3-1b-pt	1.0B	B	32K	1.0	Mar 21, 2025
	gemma-3-1b-it	1.0B	I	32K	1.0	Apr 4, 2025
	gemma-3-4b-pt	4B	B	128K	1.0	Mar 21, 2025
	gemma-3-4b-it	4B	I	128K	1.0	Mar 21, 2025
Meta	Llama-3-8B-Instruct	8B	I	8K	0.6	Jun 18, 2025
	Llama-3.1-8B-Instruct	8B	I	128K	0.6	Sep 25, 2024
	Llama-3.1-8B	8B	B	128K	0.6	Oct 16, 2024
	Llama-3.2-3B-Instruct	3B	I	128K	0.6	Oct 24, 2024
Qwen	Qwen2.5-1.5B	1.5B	B	32K	1.0	Oct 8, 2024
	Qwen2.5-7B	7B	B	128K	1.0	Sep 25, 2024
	Qwen2.5-7B-Instruct	7B	I	128K	0.7	Jan 12, 2025
	Qwen2.5-14B-Instruct	14B	I	128K	0.7	Sep 25, 2024
	Qwen3-8B	8B	B	32K	0.6	Jul 26, 2025
Mistral AI	Mistral-7B-Instruct-v0.1	7B	I	8K	1.0	Jul 24, 2025
	Mistral-7B-Instruct-v0.2	7B	I	8K	1.0	Jul 24, 2025
	Mistral-7B-Instruct-v0.3	7B	I	8K	1.0	Jul 24, 2025
	Mistral-7B-v0.3	7B	B	8K	1.0	Jul 24, 2025
Microsoft	Phi-4-mini-instruct	4B	I	128K	1.0	May 1, 2025
	phi-4	14B	B	16K	1.0	Feb 24, 2025
	Phi-4-reasoning	14B	R	32K	0.8	Jun 13, 2025
DeepSeek AI	R1-Distill-Llama-8B	8B	D	32K	0.6	Feb 24, 2025
	R1-Distill-Qwen-7B	7B	D	32K	0.6	Feb 24, 2025
	R1-0528-Qwen3-8B	8B	D	32K	1.0	May 29, 2025
Anthropic	Sonnet-4-5	–	R	200K	0.7	Sep 29, 2025

3.2 Modified Pipeline

Several pipeline modifications were introduced relative to Study I:

- **Classifier update.** The 2023 ViT classifier was trained on handwritten letters; the 2024 classifier was trained on machine-printed capital fonts, which are visually closer to the block-structure outputs. Both are evaluated in parallel. Switching classifiers produces only marginal re-ranking (the largest similarity delta is +0.0017), confirming that the 2023 classifier's documented shortcomings [2] do not qualitatively alter cross-model comparisons.

- **Unweighted aggregate score.** The competition's dynamic per-letter weighting and normalization scheme makes scores incomparable across experiments. Study II therefore uses raw unweighted scores (Equation 1).
- **Robustness handling.** Quote-character variants (curly vs. straight quotes), missing backtick delimiters, and empty responses are explicitly logged. Empty-response counts are used as a diagnostic of model-level inference failure (e.g., Llama-3.1-8B produced only 47 valid outputs from 280 attempts).

$$\text{model\_score} = \frac{1}{K}\sum_{k=1}^{K}\frac{1}{26}\sum_{j=1}^{26}\frac{1}{n}\sum_{i=1}^{n}\text{stability}(x_{ij}^{k}) \times \sigma(z_{ij}^{k}) \quad (1)$$

3.3 Contiguous Coverage Metric

The central methodological contribution of Study II is the **contiguous coverage** metric. Given a model's generated `ab_drop()` sequence for a target letter and the original prompt text, the Longest Common Contiguous Substring (LCS) algorithm identifies the longest verbatim match between the generated code and any code segment present in the prompt. Coverage is then normalized to $[0, 1]$ by the length of the generated sequence.

High contiguous coverage (≈1.0) indicates that the model reproduced a letter's solution directly from the prompt (explicit retrieval). Low coverage indicates that the output was not directly copied, though this alone does not confirm generalization—the model may have produced malformed or wrong-letter output. The metric is therefore used in conjunction with the ViT similarity score and an LLM-judge verdict to triangulate behavior.

3.4 LLM-as-Judge

Claude Sonnet 4.5 was deployed as an auxiliary generalization judge. Given a pair of images (the prompt's example letter, if any, and the generated level), the judge produces a categorical verdict (generalized / memorized / partial memorization / failed generalization / unclear) and a brief reasoning trace. This evaluator is treated as a *judge-relative* lens rather than a ground-truth oracle: it brings higher-order visual reasoning that the ViT classifier lacks, but also its own representational biases. Disagreements between the ViT score, contiguous coverage, and the judge verdict are treated as informative rather than erroneous.

3.5 Model Performance Results

Table 2 summarizes the top model rankings under both classifiers. Several findings are notable:

Instruction tuning vs. base models. All four pure base models in the evaluation (Mistral-7B-v0.3, gemma-3-4b-pt, gemma-3-1b-pt, Qwen2.5-1.5B) rank in the bottom third (ranks 15, 17, 18, 19 respectively), with mean scores between 0.027 and 0.050. The sole exception is `phi-4` (rank 2), a 14B-parameter base model whose scale appears sufficient to compensate for the absence of instruction tuning on this task.

Reasoning-tuned models underperform. Despite competitive parameter counts, reasoning-specialized variants (Phi-4-reasoning, DeepSeek-R1 distilled series) consistently rank in the lower half. Their chain-of-thought-oriented fine-tuning appears to suppress the open-ended generative

Table 2: Top-10 model rankings by mean score (2024 classifier). Instruction-tuned models dominate; stability scores are nearly uniform across all models, so ranking is determined by similarity.

Rank	Model	Sim.	Score
1	Mistral-7B-Instruct-v0.3	0.0138	0.1034
2	phi-4 (base, 14B)	0.0154	0.1035
3	Mistral-7B-Instruct-v0.2	0.0121	0.0990
4	Qwen2.5-14B-Instruct	0.0149	0.0947
5	Qwen2.5-7B (base)	0.0097	0.0941
6	Llama-3-8B-Instruct	0.0163	0.0905
7	Llama-3.2-3B-Instruct	0.0117	0.0876
8	Qwen2.5-7B-Instruct	0.0129	0.0882
9	gemma-3-4b-it	0.0146	0.0861
10	Mistral-7B-Instruct-v0.1	0.0055	0.0828

fluency the task requires: many DeepSeek-R1 responses verbally describe the correct solution but produce no executable code.

Scale within families. Within the Qwen and Gemma families, larger and instruction-tuned variants consistently outperform smaller or base counterparts. The one notable inversion is Qwen2.5-7B (base, rank 5) outperforming Qwen2.5-7B-Instruct (rank 8), driven entirely by a higher similarity score; the authors hypothesize alignment-induced output rigidity in the instruct variant.

Mistral successive versions. The three Mistral-7B-Instruct versions show monotonically increasing similarity scores across v0.1 → v0.2 → v0.3, consistent with iterative improvements to instruction-tuning data and alignment.

3.6 Generalization vs. Memorization Analysis

The contiguous-coverage heatmap as it can be seen in Figure 4 reveals a strong positive correlation between performance rank and coverage: top-performing models retrieve explicit prompt content with high fidelity, while lower-ranked models either fail to retrieve correctly or produce out-of-distribution outputs.

Key tension. High score $\not\Rightarrow$ generalization. The top-ranked models (Mistral-7B-Instruct variants, phi-4) exhibit high contiguous coverage on prompts that contain full or partial lookup tables, meaning their scores reflect reliable retrieval and formatting rather than structural inference about letter geometry.

Exceptions and nuances. The base models gemma-3-1b-pt and gemma-3-4b-pt are ranked by the zero-shot prompt *zeilde* (which contains no explicit examples) rather than by lookup-table prompts, suggesting that for weaker models incapable of extraction, coverage drops to zero by default rather than by genuine generalization.

Judge vs. systematic metric. When Sonnet 4.5 acts as judge, the overall generalization detection rate increases relative to the ViT+LCS combination. The judge identifies Phi-4-reasoning as among the least memorizing models (consistent with LCS), while flagging DeepSeek-Qwen-7B and Qwen3-8B as the most generalizing—a divergence from the ViT-based ranking that illustrates the evaluator-dependence of "generalization" in this benchmark.

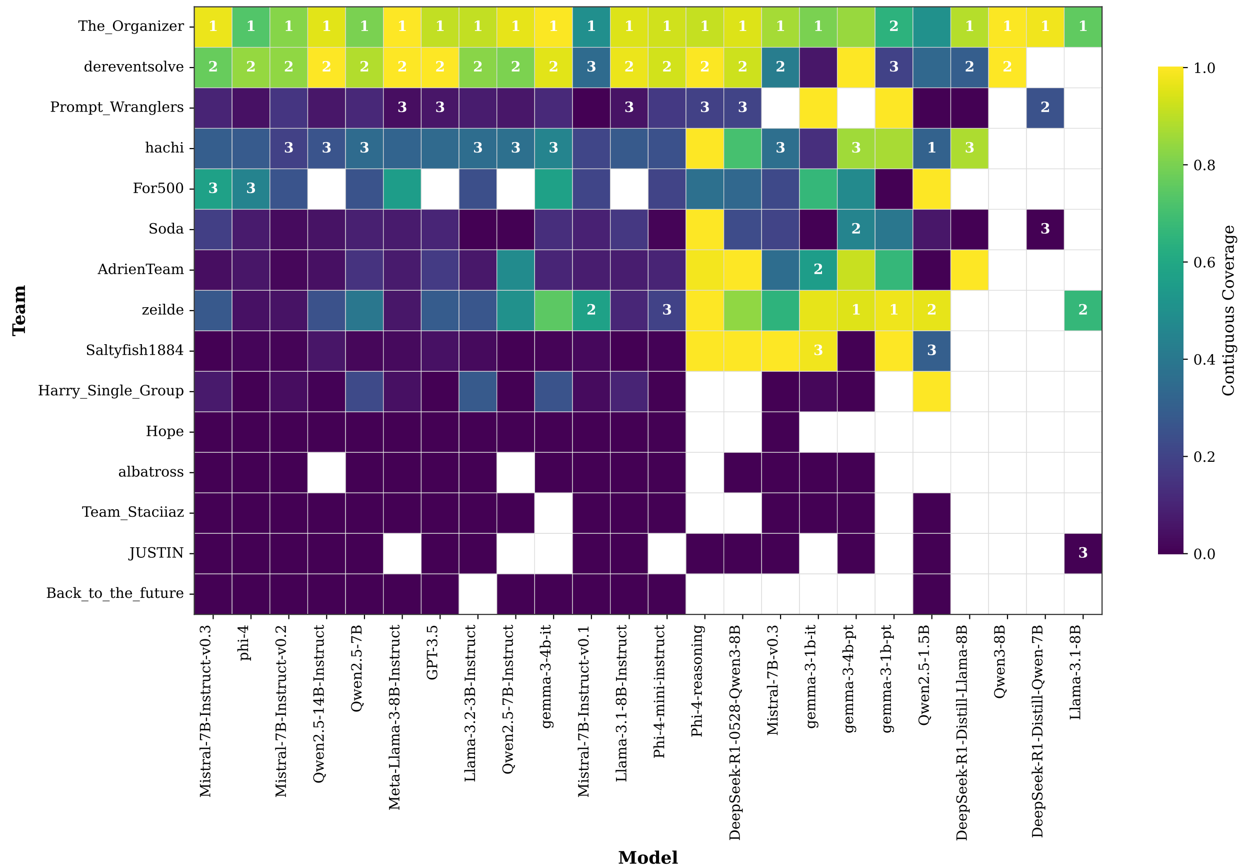


Figure 4: The Longest Common Contiguous Subsequence heatmap for all teams and models. The contiguous coverage range depicts to what extend the generated letters are taken exactly from the examples in the prompt. The top 3 rankings are based on the total scores on the teams for the respective model.

Retrieval vs. abstraction as distinct competencies. The benchmark naturally partitions into *covered* letters (explicitly in the prompt) and *uncovered* letters (not in the prompt). The covered regime tests explicit retrieval: can the model identify, select, and faithfully reproduce the correct construction? The uncovered regime tests abstractive transfer: can the model infer a valid construction from partial examples, textual rules, or analogical structure? Aggregating both regimes under a single score conflates these distinct skills.

3.7 Machine-Authored Prompt and Communicative Alignment

Sonnet 4.5 was given the competition specification and all 15 human-authored prompts and asked to produce a new prompt for itself to execute. The resulting machine-authored entry uses an 8-shot few-shot strategy with chain-of-thought decomposition. This creates a self-referential setting of *communicative alignment*: a model designs instructions optimized for another instance sharing similar representational biases. The result is treated as an initial probe of model-aware prompt design rather than a strong claim about theory of mind.

4 Interface

In order to effectively browse and analyze the approximately 100,000 output instances, a simplistic interface is designed. You can see a snapshot of the interface in Figure 5 and 6.

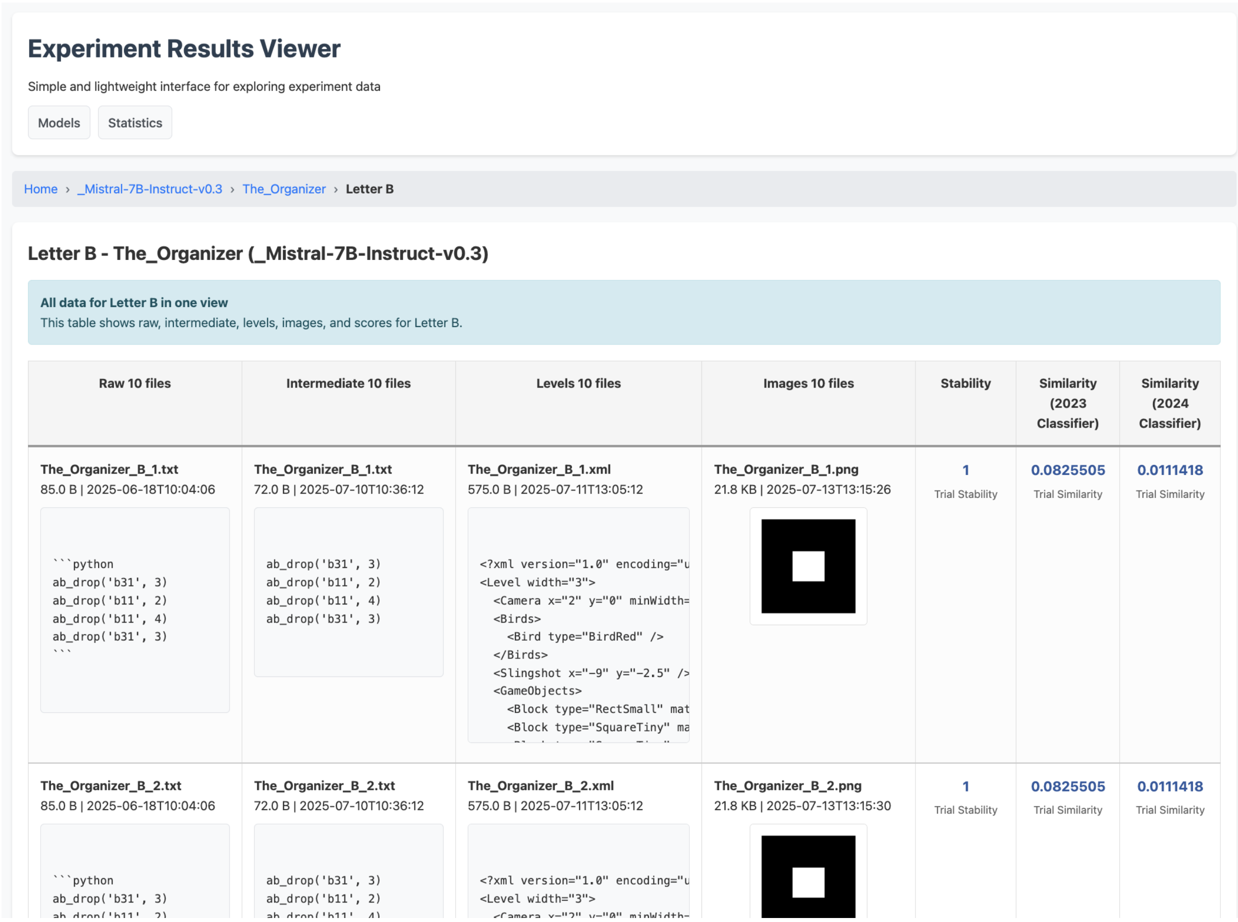
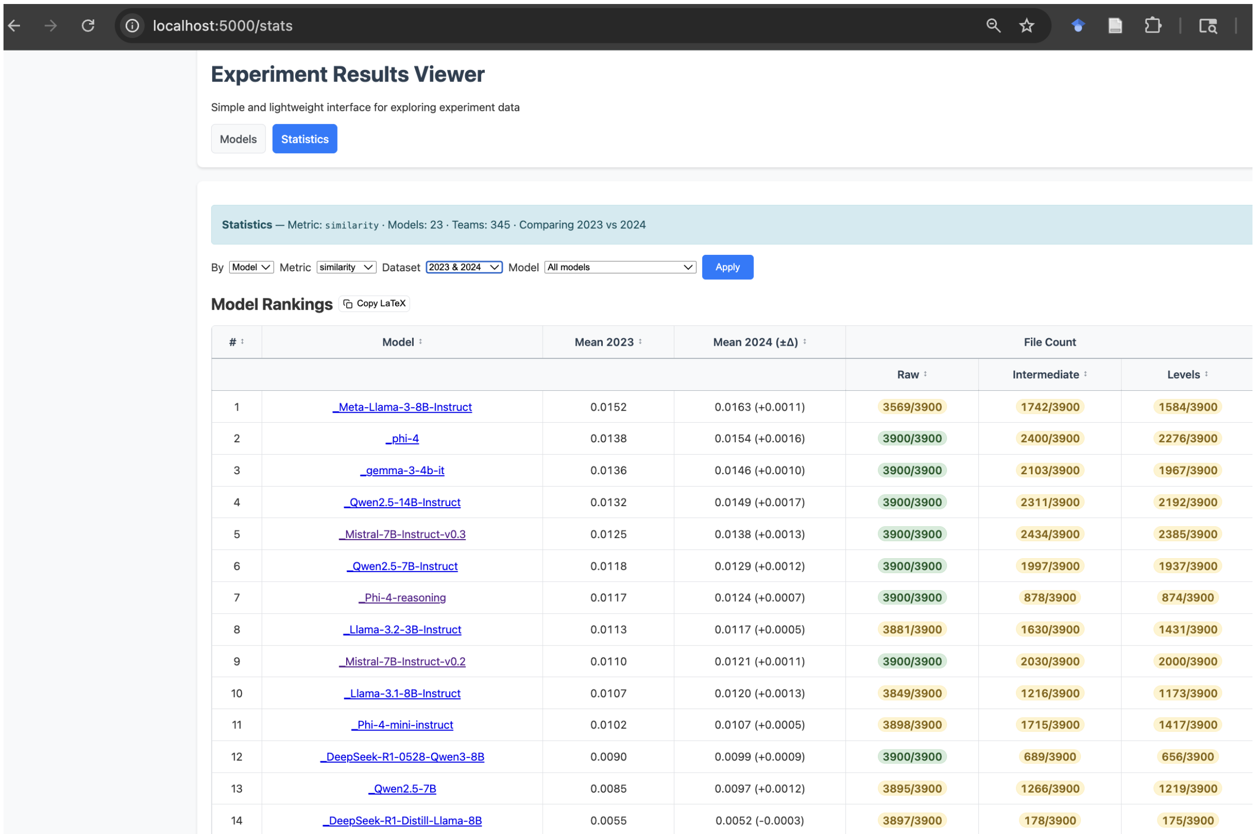


Figure 5: The basic output browser: after selecting the model, team/prompt and the letter, you can see the raw responses received back from the respective model, intermediate file (exctracted ab_drop() commands), actual .XML levels, the generated letter snapshot, stability and the similarity scores for 2023-2024 classifiers. This interface is interactive so clicking on a file preview opens up the full contents of the file in a new box.

5 Cross-Study Synthesis

The two studies, taken together, establish three conceptual claims:

1. **Replicability requires static weights.** Closed-source model updates can alter prompt scores significantly (Study I: two prompts shifted from $\sim$0 to top-5 performance), making longitudinal comparison unreliable without archiving raw model outputs.
2. **Competition rankings measure lookup-table fidelity, not generalization.** Both studies confirm, through different methods (N-shot ablation in Study I; LCS coverage in Study II), that high-scoring prompts succeed primarily by providing complete explicit solutions and that LLMs retrieve rather than generalize.
3. **Generalization is judge-relative.** The ViT classifier, the LCS metric, and the LLM judge do



#	Model	Mean 2023	Mean 2024 (±Δ)	File Count		
				Raw	Intermediate	Levels
1	_Meta-Llama-3-8B-Instruct	0.0152	0.0163 (+0.0011)	3569/3900	1742/3900	1584/3900
2	_phi-4	0.0138	0.0154 (+0.0016)	3900/3900	2400/3900	2276/3900
3	_gemma-3-4b-it	0.0136	0.0146 (+0.0010)	3900/3900	2103/3900	1967/3900
4	_Qwen2.5-14B-Instruct	0.0132	0.0149 (+0.0017)	3900/3900	2311/3900	2192/3900
5	_Mistral-7B-Instruct-v0.3	0.0125	0.0138 (+0.0013)	3900/3900	2434/3900	2385/3900
6	_Qwen2.5-7B-Instruct	0.0118	0.0129 (+0.0012)	3900/3900	1997/3900	1937/3900
7	_Phi-4-reasoning	0.0117	0.0124 (+0.0007)	3900/3900	878/3900	874/3900
8	_Llama-3.2-3B-Instruct	0.0113	0.0117 (+0.0005)	3881/3900	1630/3900	1431/3900
9	_Mistral-7B-Instruct-v0.2	0.0110	0.0121 (+0.0011)	3900/3900	2030/3900	2000/3900
10	_Llama-3.1-8B-Instruct	0.0107	0.0120 (+0.0013)	3849/3900	1216/3900	1173/3900
11	_Phi-4-mini-instruct	0.0102	0.0107 (+0.0005)	3898/3900	1715/3900	1417/3900
12	_DeepSeek-R1-0528-Qwen3-8B	0.0090	0.0099 (+0.0009)	3900/3900	689/3900	656/3900
13	_Qwen2.5-7B	0.0085	0.0097 (+0.0012)	3895/3900	1266/3900	1219/3900
14	_DeepSeek-R1-Distill-Llama-8B	0.0055	0.0052 (-0.0003)	3897/3900	178/3900	175/3900

Figure 6: The statistics page which lets the user to apply different filters and selections to view the various rankings. Each row of the table, depending on the selections, depicts model/team name, various scores along with the count of files generated at each step of the pipeline. These are also color coded with green, yellow and red to highlight full, partial and empty categories.

not measure the same property. Treating any single evaluator as ground truth misrepresents the multi-faceted nature of generalization in this setting. The benchmark is better understood as a laboratory for disentangling retrieval, formatting reliability, instruction following, and abstractive transfer—not as a single scalar measure of intelligence.

Across both studies, the operative discriminator between model rankings is **similarity score**, not stability. Most models produce structurally stable levels, but correctly shaped letters—whether by retrieval or generation—are rare. This implies that the bottleneck for improvement lies in letter-shape understanding rather than physics compliance.

6 Limitations

- **Classifier quality.** Both the 2023 and 2024 ViT classifiers exhibit misclassification artifacts (e.g., block-letter A classified as H with 0.99 probability). Scores are therefore approximate proxies for visual fidelity. We included CLIP vision transfer models to provide more classifications to tackle this limitation.
- **Hardware constraints.** The largest model evaluated is 14B parameters. Performance at 70B+ scale is unknown. We ran all of the open-source models on the same hardware (NJIT HPC). For GPT 3.5 and Claude Sonnet 4.5, we used the official API calls.

- **Coverage metric threshold.** The contiguous coverage metric uses a binary threshold for labeling outputs as copied vs. derived; the choice of threshold is not formally calibrated. Be that as it may, we employed Sonnet 4.5 as an bigger model judge to shed some light on copied vs. derived concept.
- **LLM judge reliability.** Sonnet 4.5 as judge introduces its own biases. Inter-rater reliability between the judge and human annotators has not been measured. Maybe a human study can be a future avenue of research.
- **Prompt authorship confound.** The 15 human prompts were designed specifically for GPT-3.5. Transfer to other models may be confounded by model-specific formatting conventions embedded in the prompts.

7 Future Work

These are some of the possible future avenues that can be explored.

DIRECTION 1

Further refinement on the pipeline. The original competition does not reward generalization specifically. Maybe including a new metric score focused on abstraction/generalization helps improve the findings. In addition, training a exclusive letter-recognition classifier fine-tuned to the black and white pixel style English capital letters will improve overall results.

DIRECTION 2

Extending the communicative alignment study to cross-family prompting (one model writes prompts for a different family) to test whether shared representational biases are necessary for effective self-prompting.

DIRECTION 3

Explore the notion of prompt evolution. Taking cues from the previous future direction, maybe an automated system of evolving a prompt through evaluation and feedback loop can lead into a better prompting strategy that sheds light on communicative alignment.

DIRECTION 4

Designing a focused study on theory of the mind. This avenue would bring the current study and the potential 'communicative alignment' aspect under a more philosophical light.

DIRECTION 5

Formally calibrating the LCS coverage threshold via human annotation of generated levels as memorized vs. generalized.

References

[1] P. Taveekitworachai, F. Abdullah, M. F. Dewantoro, R. Thawonmas, J. Togelius, and J. Renz. *ChatGPT4PCG Competition: Character-Like Level Generation for Science Birds*. IEEE Conference on Games, 2023.

[2] A. Moradi Karkaj, M. J. Nelson, I. Koutis, and A. K. Hoover. *Prompt Wrangling: On Replication and Generalization in Large Language Models for PCG Levels*. Foundations of Digital Games (FDG), 2024.

[3] A. Summerville *et al. Procedural Content Generation via Machine Learning (PCGML)*. IEEE Transactions on Games, 10(3):257–270, 2018.